

do49_classification_linear-models

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Linear Models

Classifiers that fit a linear (or L1-regularized linear) decision boundary through the Generalized Linear Model framework — the family every other classification deck in this collection assumes the reader already recognizes but never formally introduces.

2 TECHNIQUES • CLA_GLM • CLA_GLMNET

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Where Linear Models Fit

Classification methods can be mapped into families: tree-based, probabilistic, distance-based, margin-based, ensemble — and linear models. A linear model predicts class membership through a weighted sum of the input attributes passed through a link function, so the decision boundary is a hyperplane in feature space.

- **Simplicity as a feature, not a limitation** — a linear boundary is easy to inspect, easy to explain to a non-technical stakeholder, and hard to overfit on small datasets.
- **The natural first alternative to a tree** — where `cla_dtree` splits the space into axis-aligned regions, a GLM draws one smooth separating surface.

- **Foundation for everything regularized** — L1/L2 penalties, elastic nets, and even the linear layer inside a neural network all extend the same idea.

Logistic Regression — `cla_glm`

Binary logistic regression through R's Generalized Linear Model framework. The target must be binary; the log-odds of the positive class are modeled as a linear combination of the predictors.

- Requires a binary target — the `daltoolbox` example remaps the three-class `iris$Species` into `versicolor` vs. `not_versicolor` to fit this requirement.
- Same Experiment Line as every other classifier: `cla_glm(target, positive = ...)` → `fit()` → `predict()` → `evaluate()`.
- Coefficients are directly interpretable as log-odds contributions — a rare property among the classifiers in this collection.

```
model <- cla_glm("IsVersicolor", positive = "versicolor")
model <- fit(model, iris_train)
prediction <- predict(model, iris_test)
evaluate(model, iris_test$IsVersicolor, prediction)
```

Reference: McCullagh, P., and Nelder, J. A. (1989). *Generalized Linear Models*.

L1-Regularized Logistic Regression — `cla_glmnet`

Keeps the exact same binary classification line of experiment as `cla_glm`, but replaces ordinary maximum-likelihood fitting with L1-penalized (lasso) fitting, which can shrink irrelevant coefficients to exactly zero.

- Same workflow, one extra knob: `lambda` controls how aggressively coefficients are penalized.
- `lambda = "lambda.1se"` picks the most regularized model within one standard error of the best cross-validated fit — a common, conservative default.
- Useful when the predictor set is large or collinear and a plain GLM would overfit or become numerically unstable.

```
model <- cla_glmnet("IsVersicolor", lambda = "lambda.1se")
model <- fit(model, iris_train)
```

Reference: Friedman, J., Hastie, T., and Tibshirani, R. (2010). *Regularization Paths for Generalized Linear Models via Coordinate Descent*.

What to Observe

- Both examples keep the exact same experiment body used across the whole classification collection — only the learner and the target's binary requirement change.
- `cla_glm` favors interpretability; `cla_glmnet` favors robustness under many or correlated predictors.
- This family is the linear counterpart to every non-linear method covered in the other decks (kNN, SVM, trees, ensembles, neural networks) — worth presenting as the baseline they are all compared against.